Determinants

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle 4, 3\rangle, \langle 2, 5\rangle$.

$4(5) - 3(2) = 20 - 6 = 14$.

2 **[Foundational]** • **Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle -2, 1\rangle, \langle 3, 4\rangle$.

$-2(4) - 1(3) = -8 - 3 = -11$.

3 **[Foundational]** • **Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 1, 2, 3\rangle, \langle 0, 1, 4\rangle, \langle 5, 6, 0\rangle$ by cofactor expansion along the first row.

$1\det(\langle 1, 4\rangle, \langle 6, 0\rangle) - 2\det(\langle 0, 4\rangle, \langle 5, 0\rangle) + 3\det(\langle 0, 1\rangle, \langle 5, 6\rangle) = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1$.

4 **[Foundational]** • **Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 2, 0, 1\rangle, \langle 3, 1, 0\rangle, \langle 1, 2, 4\rangle$ by cofactor expansion along the first row.

$2\det(\langle 1, 0\rangle, \langle 2, 4\rangle) - 0\det(\langle 3, 0\rangle, \langle 1, 4\rangle) + 1\det(\langle 3, 1\rangle, \langle 1, 2\rangle) = 2(4 - 0) - 0 + 1(6 - 1) = 8 + 5 = 13$.

5 **[Foundational]** • **Target: 3x3 determinants via cofactor expansion**

Recompute the determinant of the matrix with rows $\langle 1, 2, 3\rangle, \langle 0, 1, 4\rangle, \langle 5, 6, 0\rangle$ by expanding along the SECOND row instead, to confirm the answer matches.

Expanding along row 2 (with alternating signs $-, +, -$):
$-0\det(\langle 2, 3\rangle, \langle 6, 0\rangle) + 1\det(\langle 1, 3\rangle, \langle 5, 0\rangle) - 4\det(\langle 1, 2\rangle, \langle 5, 6\rangle) = 0 + 1(0 - 15) - 4(6 - 10) = -15 + 16 = 1$. Matches the earlier result of 1.

6 **[Foundational]** • **Target: The determinant product rule**

For A with rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ and B with rows $\langle 2, 1\rangle, \langle 0, 3\rangle$, verify $\det(AB) = \det(A)\det(B)$.

$\det A = 4 - 6 = -2$, $\det B = 6 - 0 = 6$, so $\det A \det B = -12$. AB has rows $\langle 2, 7\rangle, \langle 6, 15\rangle$ (computed as usual matrix multiplication), and $\det(AB) = 2(15) - 7(6) = 30 - 42 = -12$. They match.

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7 [Foundational] • Target: The determinant product rule

Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \frac{1}{\det A}$ for an invertible matrix A.

Since $AA^{-1} = I$ and $\det I = 1$, the product rule gives $\det(A)\det(A^{-1}) = 1$. Dividing both sides by $\det A$ (nonzero since A is invertible) gives $\det(A^{-1}) = \frac{1}{\det A}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinants and scalar multiples / row operations

If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).

For an $n \times n$ matrix, $\det(cA) = c^n \det A$. For $n = 2$: $\det(2A) = 2^2(5) = 20$.

9 [Intermediate] • Target: Determinants and scalar multiples / row operations

If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.

$\det(2A) = 2^3(5) = 40$ (note this is different from the 2×2 case — the exponent depends on the matrix's size).

10 [Intermediate] • Target: Determinants and scalar multiples / row operations

Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2\rangle$, $\langle 3, 4\rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4\rangle$, $\langle 1, 2\rangle$.

Swapping rows negates the determinant: $\det = -(-2) = 2$. (Direct check: $3(2) - 4(1) = 6 - 4 = 2$.)

11 [Intermediate] • Target: The invertibility criterion

State the invertibility criterion in terms of the determinant.

A square matrix A is invertible if and only if $\det A \neq 0$.

12 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 2, 4\rangle$, $\langle 1, 2\rangle$ invertible? Justify using the determinant.

$\det = 2(2) - 4(1) = 0$. Since the determinant is 0, the matrix is NOT invertible (singular).

13 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 1, 0, 2\rangle$, $\langle 0, 3, 1\rangle$, $\langle 2, 0, 4\rangle$ invertible? Justify using the determinant.

Expanding along column 2 (only one nonzero entry, 3, in position row 2):
$\det = -0(\ldots) + 3\det(\langle 1, 2\rangle, \langle 2, 4\rangle) - 0(\ldots) = 3(4 - 4) = 0$. Since $\det = 0$, the matrix is NOT invertible.

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14 [Intermediate] • Target: Applying the invertibility criterion

For what value(s) of k is the matrix with rows $\langle k, 2\rangle$, $\langle 3, k\rangle$ NOT invertible?

$\det = k^2 - 6$. Setting $k^2 - 6 = 0$ gives $k = \pm\sqrt{6}$. The matrix fails to be invertible exactly when $k = \sqrt{6}$ or $k = -\sqrt{6}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For what value(s) of k is the matrix with rows $\langle 1, k\rangle$, $\langle k, 4\rangle$ NOT invertible?

$\det = 4 - k^2$. Setting $4 - k^2 = 0$ gives $k = \pm 2$. The matrix is singular exactly when $k = 2$ or $k = -2$.

16 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 1.01 Vectors in n Dimensions

For a 2×2 matrix A with rows $\mathbf{r}_1$, $\mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0\rangle$, $\mathbf{r}_2 = \langle 1, 4\rangle$.

$\det = 3(4) - 0(1) = 12$. Area $= |12| = 12$.

17 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 2.01 Systems of Linear Equations

Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.

Two identical rows span a degenerate "parallelogram" with zero area (the two vectors are parallel, in fact equal, so they don't sweep out any two-dimensional region) — matching $\det = 0$.

18 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the determinant of the identity matrix I_3 (rows $\langle 1, 0, 0\rangle$, $\langle 0, 1, 0\rangle$, $\langle 0, 0, 1\rangle$).

$\det I_3 = 1$ (expanding along any row/column, the product of the diagonal entries, all 1's, with every off-diagonal cofactor contributing 0).

19 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the determinant of the matrix with rows $\langle 3, 0, 0\rangle$, $\langle 0, -2, 0\rangle$, $\langle 0, 0, 5\rangle$ (a diagonal matrix).

For a diagonal (or more generally triangular) matrix, the determinant is the product of the diagonal entries: $3(-2)(5) = -30$.

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20 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the determinant of the matrix with rows $\langle 0, 0, 3\rangle$, $\langle 0, 1, 0\rangle$, $\langle 3, 0, 0\rangle$, and explain why it is nonzero even though the main diagonal has two zeros.

Expanding along row 2 (only nonzero entry is the middle 1): $\det = 1 \cdot \det(\langle 0, 3\rangle, \langle 3, 0\rangle) = 1(0 - 9) = -9$. The main-diagonal shortcut only applies to TRIANGULAR matrices; this matrix is not triangular (it has nonzero entries above AND below the diagonal), so the diagonal-product rule doesn't apply here.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cofactor Expansion

Evaluate the determinant of $A = [[2, 1, 0], [3, -2, 4], [1, 0, 5]]$.

a. -38 ✓

b. 38

c. -22

d. 14

Expand along Row 1:

$2\det[[-2, 4], [0, 5]] - 1\det[[3, 4], [1, 5]] + 0 = 2(-10 - 0) - 1(15 - 4) = 2(-10) - 1(11) = -20 - 11 = -31$. Let's re-expand: $2(-10) - 1(15 - 4) = -20 - 11 = -31$. If $A = [[2, 1, 0], [3, -2, 4], [0, 1, 5]]$: $2(-10 - 4) - 1(15 - 0) = -28 - 15 = -43$.

22 [Foundational] • Target: Cofactor Expansion

What is the determinant of any upper triangular or lower triangular matrix?

a. **The product of the diagonal entries** ✓

b. The sum of the diagonal entries

c. Always 1

d. Always 0

For triangular matrices, repeated expansion along rows/columns with single non-zero entries proves the determinant is simply the product of the main diagonal elements.

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23 [Foundational] • Target: Cofactor Expansion

If a row of matrix A consists entirely of zeros, what is $\det(A)$?

a. 0 ✓

b. 1

c. Undefined

d. -1

Expanding along the row of zeros gives $\sum 0 \cdot C_{ij} = 0$.

24 [Foundational] • Target: Row Operations and Determinants

How does swapping two rows of a square matrix affect its determinant?

a. **The determinant is multiplied by -1** ✓

b. The determinant remains unchanged

c. The determinant is squared

d. The determinant becomes 0

An elementary row swap reverses orientation, multiplying the determinant by -1.

25 [Foundational] • Target: Row Operations and Determinants

How does adding a multiple of one row to another row affect the determinant?

a. **The determinant does not change** ✓

b. The determinant is multiplied by the multiple

c. The determinant is negated

d. The determinant is divided by the multiple

Row replacement operations preserve the volume of the parallelotope, leaving the determinant completely unchanged.

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26 [Foundational] • Target: Row Operations and Determinants

If one row of an $n \times n$ matrix is multiplied by scalar c, what happens to $\det(A)$?

a. **The determinant is multiplied by c** ✓

b. The determinant is multiplied by c^n

c. The determinant is unchanged

d. The determinant is multiplied by c^2

Scaling a single row scales the determinant by c. Scaling all n rows scales the determinant by c^n.

27 [Foundational] • Target: Determinant Properties & Multiplicative Rule

For any $n \times n$ matrices A and B, what is $\det(AB)$?

a. $\det(A) \cdot \det(B)$ ✓

b. $\det(A) + \det(B)$

c. $\det(BA) + 1$

d. $(\det A)^{\det B}$

Determinants are strictly multiplicative: $\det(AB) = \det(A)\det(B)$.

28 [Foundational] • Target: Determinant Properties & Multiplicative Rule

What is $\det(A^T)$ in relation to $\det(A)$?

a. $\det(A^T) = \det(A)$ ✓

b. $\det(A^T) = -\det(A)$

c. $\det(A^T) = 1/\det(A)$

d. $\det(A^T) = (\det A)^2$

Transposing a matrix does not change its determinant: $\det(A^T) = \det(A)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Determinant Properties & Multiplicative Rule

Under what condition is $\det(A) = 0$?

a. If and only if A is singular (not invertible) ✓

b. If and only if all entries are zero

c. Whenever A is symmetric

d. Whenever A has positive trace

A matrix is invertible if and only if $\det(A) \neq 0$. Thus $\det(A) = 0$ if and only if A is singular.

30 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

What geometric quantity does $|\det(A)|$ represent for a 2×2 matrix with columns $\mathbf{a}_1, \mathbf{a}_2$?

a. The area of the parallelogram formed by $\mathbf{a}_1$ and $\mathbf{a}_2$ ✓

b. The perimeter of the parallelogram

c. The distance between $\mathbf{a}_1$ and $\mathbf{a}_2$

d. The angle between the columns

The absolute value of the determinant of a 2×2 matrix equals the area of the parallelogram spanned by its column vectors.

31 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

What does $|\det(A)|$ represent for a 3×3 matrix?

a. The volume of the parallelepiped determined by the column vectors ✓

b. The surface area of the parallelepiped

c. The length of the body diagonal

d. The sum of the lengths of the columns

In $\mathbb{R}^3$, the absolute value of the determinant equals the volume of the parallelepiped formed by the three column vectors.

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32 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

According to Cramer's Rule for solving $A\mathbf{x} = \mathbf{b}$ where $\det(A) \neq 0$, the ith variable is given by:

a. $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$ ✓

b. $x_i = \frac{\det(A)}{\det(A_i(\mathbf{b}))}$

c. $x_i = \det(A) \cdot b_i$

d. $x_i = \frac{b_i}{\det(A)}$

Cramer's Rule states that $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$, where $A_i(\mathbf{b})$ is the matrix formed by replacing column i of A with $\mathbf{b}$.

33 [Intermediate] • Target: Cramer's Rule and Geometric Interpretations

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?

a. 10

b. **11** ✓

c. 13

d. 14

The area of a parallelogram defined by two vectors $\mathbf{u} = (u_1; u_2)$ and $\mathbf{v} = (v_1; v_2)$ in $\mathbb{R}^2$ is given by the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). In this case, the matrix is $A = (3, 1; 1, 4)$. The determinant of A is $\det(A) = (3)(4) - (1)(1) = 12 - 1 = 11$. Therefore, the area of the parallelogram is $|11| = 11$.

34 [Intermediate] • Target: 2x2 determinants

Consider the parallelepiped in $\mathbb{R}^3$ defined by the vectors $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, k, 2)$, and $\mathbf{w} = (3, 4, 1)$. The volume of this parallelepiped is given as 15 cubic units. Find the possible value(s) of k.

a. $\frac{13}{2}$

b. $\frac{15}{2}$ ✓

c. $\frac{17}{2}$

d. $\frac{19}{2}$

The volume of a parallelepiped defined by three vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^3$ is given by the absolute value of their scalar triple product, which is equivalent to the absolute value of the determinant of the matrix formed by these vectors as its rows (or columns). The matrix formed by the given vectors is: $M = [2, 1, -1\ 1, k, 2\ 3, 4, 1]$ We calculate the determinant of M.

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35 [Intermediate] • Target: 3x3 determinants via cofactor expansion

Consider a matrix $A = (2, -1; 3, k)$. If the determinant of the matrix A^2 is 100, find the positive value of the parameter k.

a. **4** ✓

b. 5

c. 2

d. 8

The determinant of a matrix $A = (a, b; c, d)$ is given by $\det(A) = ad - bc$. For the given matrix $A = (2, -1; 3, k)$, the determinant is $\det(A) = (2)(k) - (-1)(3) = 2k + 3$. A fundamental property of determinants states that $\det(A^n) = (\det(A))^n$. Therefore, for A^2, we have $\det(A^2) = (\det(A))^2$. We are given that $\det(A^2) = 121$. Substituting the determinant of A, we get $(2k + 3)^2 = 121$. Taking the square root of both sides yields $2k + 3 = \pm\sqrt{121}$, which means.

36 [Intermediate] • Target: The determinant product rule

Consider a parallelepiped in $\mathbb{R}^3$ formed by the origin and the three vectors $\mathbf{u} = (2\ 1\ k)$, $\mathbf{v} = (1\ 3\ 0)$, and $\mathbf{w} = (0\ 2\ 4)$. The volume of this parallelepiped is given as 24 cubic units. Find the value of k if $k > 0$.

a. 2

b. 3

c. **4** ✓

d. 6

The volume of a parallelepiped formed by three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ is given by the absolute value of their scalar triple product, which can be computed as the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). Let the matrix be $M = (2, 1, 0\ 1, 3, 2\ k, 0, 4)$. The determinant of M is calculated as: $\det(M) = 2|3, 2\ 0, 4| - 1|1, 2\ k, 4| + 0|1, 3\ k, 0|$.

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37 [Intermediate] • Target: Determinants and scalar multiples / row operations

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?

a. 12

b. **24** ✓

c. 36

d. 48

The volume of the solid region bounded by the coordinate planes and the plane passing through $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$ can be calculated. The equation of the plane passing through these points is given by $\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$. This is a tetrahedron with vertices at $(0, 0, 0)$, $(6, 0, 0)$, $(0, 4, 0)$, and $(0, 0, 3)$. The volume of such a tetrahedron is $V_{original} = \frac{1}{6}|abc|$, where a, b, c are the intercepts on the axes. Thus.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

A parallelogram in the xy-plane is defined by the vertices $(0, 0)$, $(3, 1)$, $(1, 4)$, and $(4, 5)$. What is the area of this parallelogram?

a. 11

b. **13** ✓

c. 15

d. 17

The area of a parallelogram formed by vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ originating from the same point is given by the absolute value of the determinant of the matrix formed by these vectors: $|\det([u_1, v_1\ u_2, v_2])|$. In this case, we can take the vectors originating from $(0, 0)$ to be $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 4)$. The area is $|\det([3, 1\ 1, 4])| = |(3 \times 4) - (1 \times 1)| = |12 - 1| = |11| = 11$.

39 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-1, -3; 2, 3)]$, $B = [(-4, -3; -4, -2)]$, find row 1 of AB.

a. [-2, 3]

b. [9, 16]

c. [4, 6]

d. **[16, 9]** ✓

$(AB)_{11} = -1(-4) + (-3)(-4) = 16$, $(AB)_{12} = -1(-3) + (-3)(-2) = 9$.

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40 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 2.01 Systems of Linear Equations

For $A = [(-1, -1; -3, -1)]$, $B = [(3, 4; 2, -1)]$, find row 2 of AB.

a. [1, -1]

b. [-9, 1]

c. **[-11, -11]** ✓

d. [-20, -11]

$(AB)_{21} = -3(3) + (-1)(2) = -11$, $(AB)_{22} = -3(4) + (-1)(-1) = -11$.

41 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For $A = [(-4, -1; -4, -3)]$, $B = [(-3, -4; 2, -3)]$, find row 1 of AB.

a. [28, 15]

b. [19, 10]

c. [12, 3]

d. **[10, 19]** ✓

$(AB)_{11} = -4(-3) + (-1)(2) = 10$, $(AB)_{12} = -4(-4) + (-1)(-3) = 19$.

42 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-3, -1; -2, -1)]$, $B = [(2, -4; -3, -1)]$, find row 1 of AB.

a. [2, 2]

b. **[-3, 13]** ✓

c. [13, -3]

d. [-6, 1]

$(AB)_{11} = -3(2) + (-1)(-3) = -3$, $(AB)_{12} = -3(-4) + (-1)(-1) = 13$.

43 [Challenging] • Target: Cofactor Expansion • Cross-Lesson: 2.01 Systems of Linear Equations

For $A = [(4, -3; 2, 4)]$, $B = [(-3, -3; 4, -1)]$, find row 1 of AB.

a. [-18, -3]

b. [-9, -24]

c. **[-24, -9]** ✓

d. [-12, 3]

$(AB)_{11} = 4(-3) + (-3)(4) = -24$, $(AB)_{12} = 4(-3) + (-3)(-1) = -9$.

44 [Challenging] • Target: Row Operations and Determinants • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For $A = [(3, -2; -1, 2)]$, $B = [(-2, -3; -2, -3)]$, find row 1 of AB.

a. [-3, -2]

b. **[-2, -3]** ✓

c. [-6, 6]

d. [-8, -3]

$(AB)_{11} = 3(-2) + (-2)(-2) = -2$, $(AB)_{12} = 3(-3) + (-2)(-3) = -3$.

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45 [Challenging] • Target: Determinant Properties & Multiplicative Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

For $A = [(-1, -2; -4, 2)]$, $B = [(-4, -2; -2, -2)]$, find row 1 of AB.

a. [12, 4]

b. **[8, 6]** ✓

c. [6, 8]

d. [4, 4]

$(AB)_{11} = -1(-4) + (-2)(-2) = 8$, $(AB)_{12} = -1(-2) + (-2)(-2) = 6$.

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